MISIAN (Minimally Invasive Surgery for Treatment of Unruptured Intracranial Aneurysms): A Prospective Randomized Single-Center Clinical Trial With Long-Term Follow-Up Comparing Different Minimally Invasive Surgery Techniques with Standard Open Surgery

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■ BACKGROUND: **Unruptured intracranial aneurysms (UIAs) are increasingly diagnosed but treatment is still controversial. Although the descriptions and use of minimally invasive surgery (MIS) have increased, comparative studies with standard approaches are rare.**

■ OBJECTIVE: **MISIAN (Minimally Invasive Surgery for Treatment of Unruptured Intracranial Aneurysms) is a prospective randomized single-center clinical trial with long-term follow-up comparing different MIS techniques with standard open surgery for treatment of UIAs.**

■ METHODS: **We randomly allocated a standard pterional approach (PtA) or MIS (1:2) to 111 patients with UIAs of the anterior circulation (mean dome diameter, 6.4 mm; range, 3–20 mm). Patients selected for MIS underwent a second randomization between a transeyelid approach (TeIA) or nanopterional approach (NPtA) (1:1).**

■ RESULTS: **Forty-one patients were randomized to and treated with the PtA, 36 with the TeIA, and 34 with the NPtA. Only patients treated with PtA had permanent facial nerve palsy ($n = 4$ [10%]; $P = 0.032$). MIS cosmetic results were considered better than those of PtA by independent observers ($P < 0.001$), and less temporal atrophy in the MIS group was also observed ($P = 0.0034$). The proportion of excellent results was higher in the TeIA group than in the NPtA group (86% vs. 67.6%; $P = 0.039$). Patients undergoing MIS also reported consistently higher satisfaction and quality-of-life scores ($P < 0.001$).**

■ CONCLUSIONS: **MIS is superior to standard PtA for microsurgical clipping of small UIAs of the anterior circulation in terms of cosmetic, satisfaction, and quality-of-life outcomes. The TeIA or NPtA for UIAs did not show significant outcome differences at 12–18 months.**

INTRODUCTION

Unruptured intracranial aneurysms (UIAs) are increasingly diagnosed,[1] with an estimated prevalence of 2%–3%.[2] Although surgical clipping is still considered the definitive form of treatment, it is being gradually replaced by less invasive endovascular treatments.[3] Despite an apparent literature support from ISAT (International Subarachnoid Aneurysm Trial),[4] results for ruptured aneurysms do not apply to UIAs. However, it is unquestionable that patients are looking for less invasive solutions.

Key words
- Aneurysm clipping
- Minimally invasive neurosurgery
- Minipterional approach
- Pterional approach
- Transeyelid approach
- Unruptured intracranial aneurysm

Abbreviations and Acronyms
MIS: Minimally invasive surgery
MISIAN: Minimally Invasive Surgery for Treatment of Unruptured Intracranial Aneurysms
mRS: Modified Rankin Scale
NPtA: Nanopterional approach
PtA: Pterional approach
TeIA: Transeyelid approach
UIA: Unruptured intracranial aneurysm
VAS: Visual analog scale
WHOQOL-Bref: World Health Organization Quality of Life Abbreviated score

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Citation: World Neurosurg. (2021) 151:e533-e544.
https://doi.org/10.1016/j.wneu.2021.04.079

*Journal homepage: **www.journals.elsevier.com/world-neurosurgery***

*Available online: **www.sciencedirect.com***

Motivated by this context, a prospective randomized controlled trial comparing minimally invasive surgery (MIS) versus standard open surgery for treatment of UIAs of the anterior circulation with the aim to evaluate clinical, surgical, and cosmetic results, patient satisfaction, and quality of life was performed at our institution. To the best of our knowledge, we report the first randomized trial comparing the use of the transeyelid approach (TelA), nanopterional approach (NPtA), and standard pterional approach (PtA).

METHODS

MISIAN (Minimally Invasive Surgery for Treatment of Unruptured Intracranial Aneurysms) is an investigator-led pragmatic single-center randomized parallel-group trial conducted in São Paulo, Brazil. The MISIAN protocol (ClinicalTrials.govNCT02345395) was approved by the institutional ethics committee. Between September 2013 and January 2017, enrollment was completed with entry of 111 patients, based on inclusion and exclusion criteria shown in Table 1. Informed consent was obtained in all patients enrolled in the study. Patients were informed that the researchers would not open quality-of-life and satisfaction assessments until completion of the study.

Patients were randomly allocated into 2 groups: MIS study group or standard PtA control group in a 2:1 ratio. Additional subgroup randomization was performed in the patients in the MIS group and patients were randomly allocated to TelA or NPtA in a 1:1 ratio (Figure 1 and Table 2). The first randomization took place 1 week before the procedure. The second randomization was performed on the night before the surgery when the patient was admitted to the hospital. Control group surgeries were divided equally by 4 experienced vascular neurosurgeons. Two of these neurosurgeons performed surgery in the MIS group.

We hypothesized that MIS is equally safe and effective for the treatment of small intradural saccular UIAs of the anterior circulation compared with PtA but could result in shorter operative times, improved cosmetic outcomes, and higher patient satisfaction and quality-of-life scores without increased rates of perioperative complications and incomplete aneurysm obliteration. Our primary and secondary outcomes are shown in Table 3. All patients underwent 3 outpatient visits at the following periods: 15—30 days, 3—6 months, and 12—18 months.

Surgical Technique

All patients were operated on by experienced vascular neurosurgeons. Pterional craniotomy was performed in the standard fashion.[5] The term nanopterional craniotomy[6] was used to standardize the keyhole craniotomy described by different investigators,[7-9] which is a reduced version of the minipterional craniotomy[10] (Figure 2). The TelA technique was performed as previously described by our group[11] (Figure 3). A ClearScope adapter (Clearwater Clinical, Ottawa, Canada) was used to attach a rigid rod lens neuroendoscope (Karl Storz, Tuttlingen, Germany) to a smartphone.[6] This construct was used in the MIS group but not with the PtA group because it is not a standard adjunct for UIA surgery in our institution.

Patient Satisfaction and Aesthetic Evaluation

The present study focused on patient satisfaction in 5 aspects: facial hypoesthesia, facial pain or headache, masticatory discomfort, aesthetic results, and general satisfaction with the procedure. A visual analog scale (VAS) digital questionnaire (Google Forms [Google LLC, Mountain View, California, USA]) was sent to all patients enrolled in the study after all follow-up visits. Aesthetic results were also evaluated by providing before and after photographs to 2 independent observers blinded to the surgical approach who graded the aesthetic results using the grading scale shown in Table 4.

Table 1. Inclusion and Exclusion Criteria

Inclusion Criteria	Exclusion Criteria
18—70 years old	Absence of adequate family support in the postoperative period at home
Incidental saccular UIAs of the anterior circulation	Pregnant or breastfeeding women
Aneurysms found after contralateral subarachnoid hemorrhage >6 months from trial inclusion	Major surgery <6 months
Aneurysms dome between 3 mm and 2 cm	Patients with cardiovascular, hepatic, or renal disease that demanded extended hospital stay by preoperative protocols
Patients with indication for aneurysm occlusion based on risk factors, past medical history and patient's psychological status*	Patients with coagulopathies or taking anticoagulation medication
	Fusiform UIAs

UIA, unruptured intracranial aneurysm.
*Psychological status = patient's level of anxiety and willingness to be treated.

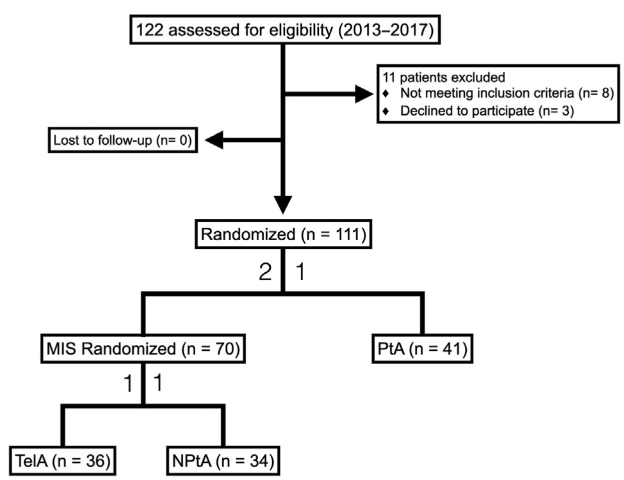

Figure 1. Study design. A total of 111 patients were randomly allocated into 2 groups, minimally invasive surgery (MIS) study group or standard pterional approach (PtA) control group, in a 2:1 ratio. Additional subgroup randomization was performed in the MIS group patients and patients were randomly allocated to the transeyelid approach (TelA) group or nanopterional approach (NPtA) group in a 1:1 ratio.

Table 2. Patient Demographics

Demographic	Transeyelid Approach Group (n = 36)	Nanopterional Approach Group (n = 34)	*P* (Transeyelid Approach × Nanopterional Approach)	Pterional Approach Group (n = 41)	Total (N = 111)	*P* (Minimally Invasive Surgery × Pterional Approach)
Age (years), Mean (SD)	57.6 (10.9)	57.3 (9.8)	0.63	56.5 (7.6)	57.1 (9.4)	0.29
Sex						
Male	26 (72.2)	29 (85.3)	0.29	33 (80.5)	88 (79.3)	0.99
Female	10 (27.8)	5 (14.7)		8 (19.5)	23 (20.7)	
Race or ethnicity (self-declaration)						
Asian	1 (2.78)	0		0	1 (0.9)	
White	28 (77.78)	25 (73.53)	0.47	25 (60.98)	78 (70.27)	0.27
Black	2 (5.56)	5 (14.71)		6 (14.63)	13 (11.71)	
Mixed race	5 (13.89)	4 (11.76)		10 (24.39)	19 (17.12)	
Diabetes mellitus	4 (11.1)	3 (8.82)	0.99	5 (12.2)	12 (10.81)	0.89
Hypertension	24 (66.67)	23 (67.65)	0.99	31 (75.61)	78 (70.27)	0.63
Hyperlipidemia	12 (33.33)	10 (29.41)	0.92	11 (26.83)	33 (29.3)	0.82
Body mass index (kg/m^2)	25.7 (3.7)	25.5 (4.5)	0.19	24.9 (3.9)	25.4 (4)	0.30
Smoking status						
Never smoked	14 (38.89)	11 (32.35)	0.85	11 (26.83)	36 (32.43)	
Stopped <2 months	4 (11.11)	4 (11.76)		8 (19.51)	16 (14.41)	0.63
Stopped >1 year	12 (33.33)	11 (32.35)		16 (39.02)	39 (35.14)	
Smoker	6 (16.67)	8 (23.53)		6 (14.63)	20 (18.02)	
Packs/year, mean (SD)	21 (23.7)	19.8 (20.8)				
American Society of Anesthesiology physical status classification system grade				20.7 (20.9)	21.4 (22.5)	
1	7 (19.44)	5 (14.71)	0.17	4 (9.76)	16 (14.41)	0.32
2	29 (80.56)	26 (76.47)		33 (80.49)	88 (79.29)	
3	0	3 (8.82)		4 (9.76)	7 (6.31)	
Aneurysm						
Diagnosis						
Purely incidental	29 (80.56)	25 (73.53)	0.23	34 (82.93)	88 (79.28)	0.75
Contralateral subarachnoid hemorrhage >6 months	7 (19.44)	9 (26.47)		7 (17.07)	23 (20.72)	
Localization						
MCA (bifurcation)	21 (53.84)	21 (58.33)	0.27	22 (48.88)	64 (53.33)	0.71
MCA (M1)	4 (10.25)	0		2 (4.44)	6 (5)	
Anterior communicating artery	5 (12.82)	4 (11.11)		8 (17.77)	17 (14.16)	
ICA bifurcation	1 (2.56)	0		3 (6.66)	4 (3.33)	

Values are number (%) except where indicated otherwise.
SD, standard deviation; MCA, middle cerebral artery; ICA, internal carotid artery.
*Aneurysm complexity meant large and broad-based aneurysms and/or aneurysms with calcification and/or aneurysms with MCA M1 segment involvement and/or aneurysms with branches coming out of dome, and/or ophthalmic artery aneurysms requiring anterior clinoid resection for clipping.

Continues

Table 2. Continued

Demographic	Transeyelid Approach Group (n = 36)	Nanopterional Approach Group (n = 34)	*P* (Transeyelid Approach × Nanopterional Approach)	Pterional Approach Group (n = 41)	Total (N = 111)	*P* (Minimally Invasive Surgery × Pterional Approach)
ICA ophthalmic segment	2 (5.12)	1 (2.77)		2 (4.44)	5 (4.16)	
ICA posterior communicating artery segment	4 (10.25)	9 (25)		6 (13.33)	19 (15.83)	
ICA choroidal segment	2 (5.12)	1 (2.77)		2 (4.44)	5 (4.16)	
Side						
Left	18 (46.15)	16 (44.44)	0.99	18 (40)	52 (43.33)	0.73
Right	21 (53.85)	20 (55.55)		27 (60)	68 (56.66)	
Dome, diameter	6.84 (3.49)	6.55 (3.25)	0.51	6.02 (2.8)	6.43 (3.11)	0.27
Neck, size	4.1 (1.57)	3.44 (1.48)	0.023	3.82 (1.57)	3.79 (1.56)	0.65
Complexity*						
Yes	6 (15.38)	4 (11.11)	0.53	8 (17.77)	18 (15)	0.50
No	33 (84.61)	32 (88.88)		37 (82.22)	102 (85)	

Values are number (%) except where indicated otherwise.

SD, standard deviation; MCA, middle cerebral artery; ICA, internal carotid artery.

*Aneurysm complexity meant large and broad-based aneurysms and/or aneurysms with calcification and/or aneurysms with MCA M1 segment involvement and/or aneurysms with branches coming out of dome, and/or ophthalmic artery aneurysms requiring anterior clinoid resection for clipping.

Quality of Life

The World Health Organization Quality of Life Abbreviated score (WHOQOL-Bref) was used to assess the quality of life of all patients during the 3 outpatient visits.[13] The Beck Depression Inventory[14] was used as a screening method for patients with clinical depression that could cause a bias in the analysis of quality of life.

Table 3. Primary and Secondary Outcomes

Primary Outcomes	Secondary Outcomes
Perioperative mortality	Surgical time
All-cause mortality	Cosmetic outcome
Neurologic outcome (poor outcome = modified Rankin Scale score ≥2)	Degree of temporal muscle atrophy
Surgical Safety measures: cerebrospinal fluid leak, postoperative intracranial hematomas, intraoperative aneurysm bleeding, surgical infection, reoperations, and perioperative clinical complications	Postoperative facial nerve palsy
	Presence of residual aneurysm neck
	Patient satisfaction (masticatory alterations, facial numbness, pain, and overall satisfaction)
	Quality of life (World Health Organization Quality of Life Abbreviated questionnaire)

Statistical Analysis

Sample calculation was based on a recent study performed at our institution.[12] χ^2 and Fisher exact tests were used to compare categorical variables, and Wilcoxon (Mann-Whitney) tests for continuous variables. Because of the myriad of tests conducted in our study, which may lead to increased overall type 1 error, we performed a multiple comparisons adjustment method suggested by Bonferroni. *P* values <0.0016 were therefore considered statistically significant for interpretation.[15] Interobserver reliability was measured with the κ coefficient, and high concordance was present when the results were >0.7.

RESULTS

A total of 111 patients were prospectively randomized and included in the study. Forty-one patients were randomized to the PtA technique, 36 to TelA, and 34 to NPtA. There was no crossover after randomization. There were no statistically significant differences among groups regarding patients' demographics, preoperative risk scores, and aneurysm characteristics. Complete follow-up was obtained in all patients during all periods. The median follow-up time was 16 months (range, 12–18 months).

Our cohort consisted of 23 male and 88 female patients with a median age of 59 years (range, 29–70 years). Most patients had purely incidental UIAs (88/111, 79.28%), but patients presenting >6 months after a previous subarachnoid hemorrhage from a contralateral aneurysm with UIAs were also included in the study (n = 23, 20.72%). Nine patients had multiple aneurysms (5/70 in

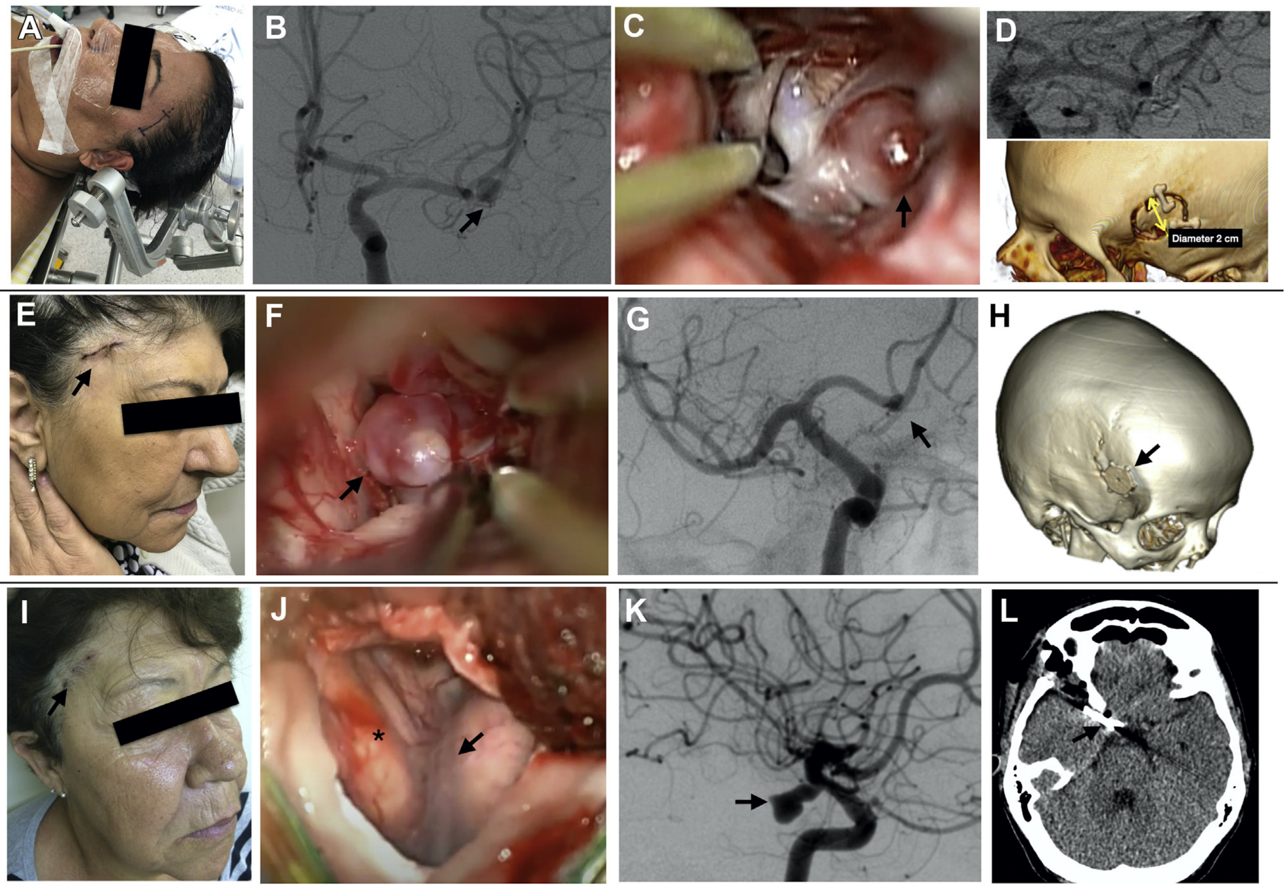

Figure 2. Nanopterional approach group case examples. Case 1: A 48-year-old woman reported chronic headaches and underwent brain imaging. (**A**) The patient is positioned supine with the head elevated 10°–15°. The neck is extended, and the head is rotated according to the aneurysm location. A straight scalp incision is marked on the hairline. The incision started at the upper temporal line, extending inferiorly 2–3 cm in the direction of the tragus. (**B**) Anteroposterior angiogram showing an aneurysm of the right middle cerebral aneurysm bifurcation (*arrow*). (**C**) Intraoperative picture demonstrating the aneurysm (*arrow*) and (**D**) postoperative angiogram showing complete aneurysm occlusion. Case 2: A 62-year-old woman who was a heavy smoker presented with a sudden headache, and she underwent brain imaging that showed a lobulated anterior communicating artery (Acom) aneurysm. No subarachnoid hemorrhage was found on the computed tomography (CT) scan or cerebrospinal fluid. (**E**) A 2.5-cm skin incision over the hairline (*arrow*). (**F**) Intraoperative image showing the lobulated Acom aneurysm (*arrow*). This image was taken with the smartphone-endoscope system to check for blind spots of the microscopic view. (**G**) Postoperative angiogram showing complete aneurysm occlusion (*arrow*). (**H**) The CT scan three-dimensional reconstruction showing the 2.2-cm craniotomy covered by a round titanium plate (*arrow*). Case 3: A 63-year-old woman reported worsening chronic headaches. A right unruptured aneurysm was shown by CT scan. (**I**) Skin incision (*arrow*). (**J**) After the incision, the muscular flap is retracted anteriorly and posteriorly, subperiosteally. A single burr hole is placed under the temporal line, just superior to the frontozygomatic suture and a 2-cm-diameter craniotomy is made with a high-speed bone drill. The sphenoid ridge is drilled until emergence of the meningo-orbital artery at the superior orbital fissure. (**J**) The dura is opened in a semilunar fashion, and the sylvian fissure can be visualized (*arrow*). (**K**) Preoperative angiogram showing the internal carotid artery aneurysm (posterior communicating artery segment). (**L**) Postoperative CT scan; the clip is indicated by the *arrow*. All patients consented to publication of their images.

the MIS group and 4/41 in the study group; P = 0.89). The aneurysm location and characteristics are further described in Table 2.

Sixty-four patients of the MIS group (91.4%) were discharged on postoperative day 1. Thirty-one PtA group patients (75.6%) were discharged 3 days after surgery as a routine institutional protocol. Reasons for extended hospital stay included stroke, cardiovascular complications, postoperative pain, and patient refusal to be discharged. The difference in delay of discharge between the MIS and PtA groups did not reach statistical significance after Bonferroni correction (P = 0.044).

Primary Outcome

One death was identified in the PtA group secondary to an intraoperative ischemic complication compared with no mortality in the MIS group (P = 0.78). At early follow-up (15–30 days), patients assigned to PtA (7/41 17.1%) trended toward poorer neurologic outcome (modified Rankin Scale [mRS] score ≥2)

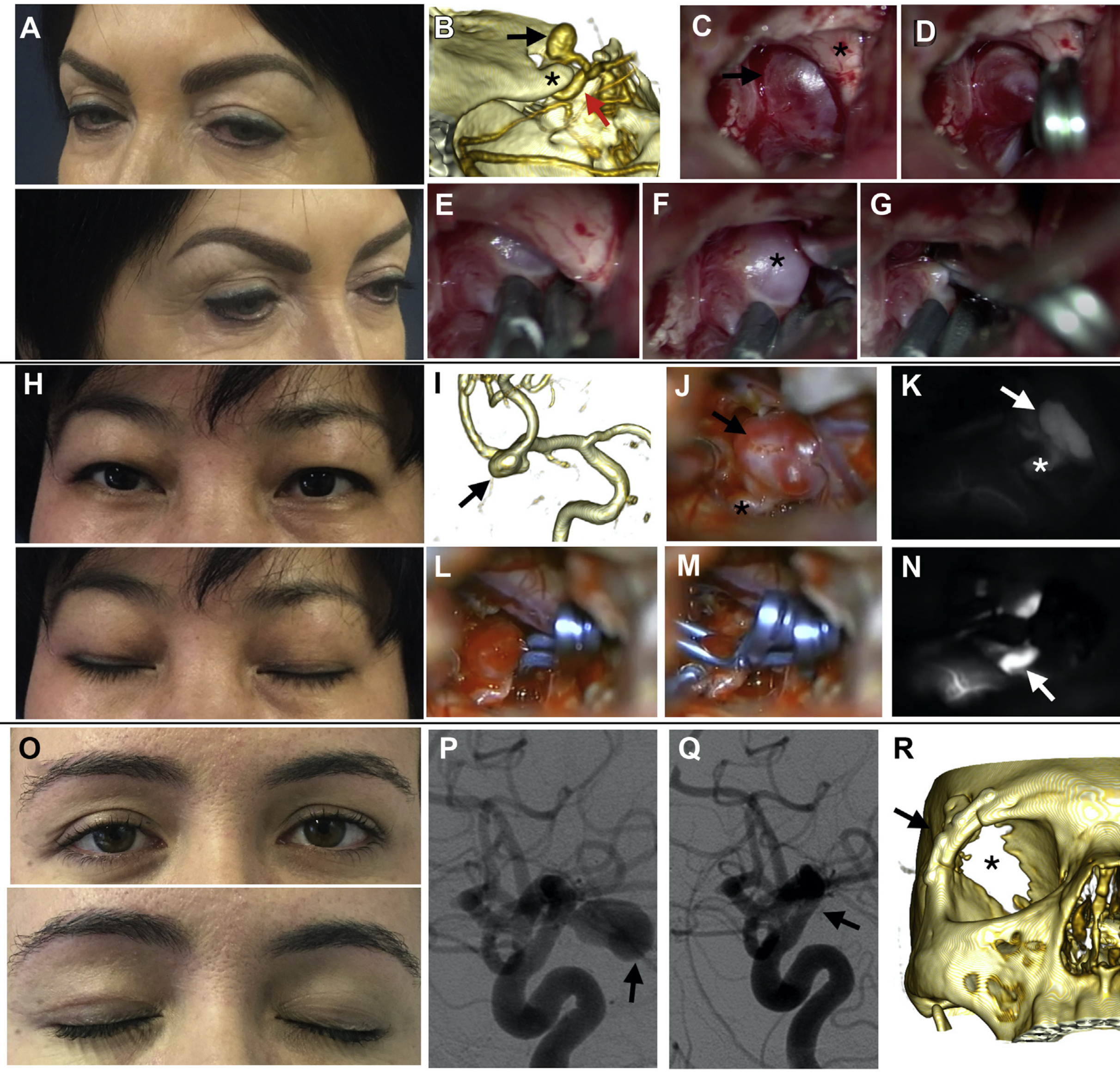

Figure 3. Transeyelid approach group case examples. Case 1: A 59-year-old woman reported blurred vision and underwent brain imaging. Preoperative neurologic examination showed a partial visual field deficit in the left eye. (**A**) Postoperative aesthetic result. (**B**) Computed tomography angiography showing a left-sided internal carotid artery (ICA) aneurysm, Ophthalmic segment (*black arrow*). *Asterisk* is over the anterior clinoid and the *red arrow* points to the cavernous segment of the ICA. Intraoperative images: (**C**) *Black arrow* indicates the aneurysm under the left optic nerve (*asterisk*). (**D**) First clip, (**E**) second clip, (**F**) third clip (*asterisk* shows the remaining aneurysm neck), and (**G**) fourth clip, with complete occlusion of the aneurysm. Case 2: A 45-year Asian woman underwent brain imaging after a head trauma. (**H**) Excellent aesthetic results can be obtained, including in a patient without an eyelid crease. (**I**) Computed tomography angiography showing a right-sided bilobulated middle cerebral artery aneurysm (*black arrow*). Intraoperative images: (**J**) After the sylvian fissure is split, the aneurysm (*arrow*) can be seen in the middle of the operative field. *Asterisk* is over one temporal branch of the middle cerebral artery. (**K**) The same findings can be observed with the intraoperative angiogram. (**L**) First and (**M**) second clips completely occluded the aneurysm as shown in the (**N**) final intraoperative angiogram. Case 3: A 33-year-old female patient with a family history of subarachnoid hemorrhage underwent brain imaging. (**O**) Postoperative aesthetic result. (**P**) Preoperative angiogram showing a right ICA aneurysm of the posterior communicating artery segment (*arrow*). (**Q**) Postoperative angiogram showing aneurysm complete occlusion (*arrow*). (**R**) Postoperative computed tomography three-dimensional reconstruction showing the craniotomy (*arrow*) and the extent of the orbital roof resection (*asterisk*). No enophthalmos was observed in any of the patients despite partial orbital roof removal. All patients consented to publication of their images.

compared with the MIS group ($P = 0.0085$). However, this finding was less pronounced at long-term follow-up (12–8 months), with 3/70 (4.2%) MIS and 6/41 (14.6%) PtA experiencing poor neurologic outcome ($P = 0.046$). There was no significant difference in neurologic outcomes within the MIS subgroups ($P = 0.09$). There were also no significant differences between the

Table 4. Results

Outcome	**Transeyelid Approach Group**	**Nanopterional Approach Group**	**P (Transeyelid Approach × Nanopterional Approach)**	**Pterional Approach Group**	**P (Minimally Invasive Surgery × Pterional Approach)**
Primary					
Mortality, number of patients	0	0	—	1 (2.44)	0.78
Neurologic Outcome (Modified Rankin Scale score ≥2)					
15—30 days	2 (5.55)	0	0.17	4 (9.75)	0.0085
3—6 months	2 (5.55)	0	0.17	4 (9.75)	0.106
12—18 months	2 (5.55)	0	0.09	3 (7.31)	0.0468
Surgical safety					
Cerebrospinal fluid leak (subcutaneous)	1 (2.77)	1 (2.94)	0.97	2 (4.87)	0.63
Strokes	2 (5.55)	0	0.378	4 (9.75)	0.077
Epidural hematomas	0	0	—	1 (2.43)	0.78
Chronic subdural	1 (2.77)	1 (2.94)	0.99	0	0.73
Intraoperative aneurysm rupture	1 (2.77)	2 (5.88)	0.95	2 (4.87)	0.99
Infections	0	0	—	1 (2.43)	0.78
Reoperations	0	0	—	2 (4.87)	0.32
Secondary					
Operative time, Mean (minutes)	212 (38)	196 (47)	0.06	293 (47)	**<0.001**
Cosmetic result					
Observer 1					
Excellent	32 (88.88)	25 (73.52)	0.0962	22 (53.65)	**0.0011**
Good	4 (11.11)	8 (23.52)		10 (24.39)	
Regular	0	1 (2.94)		5 (12.19)	
Poor	0	0		3 (7.31)	
Observer 2					
Excellent	30 (83.33)	21 (61.76)	0.0368	18 (43.90)	**<0.001**
Good	6 (16.66)	11 (32.35)		11 (26.82)	
Regular	0	2 (5.88)		8 (19.51)	
Poor	0	0		3 (7.31)	
Temporal muscle atrophy					
Grade 0	34 (94.44)	27 (79.41)	0.0687	26	0.0034
Grade I	1 (2.77)	5 (14.70)		5 (12.19)	
Grade II	1 (2.77)	2 (5.88)		5 (12.19)	
Grade III	0	0		4 (9.75)	

Values are number (%) except where indicated otherwise.

Aesthetic results: excellent (no visible scar; no atrophy; no keloid), good (partially visible scar; mild atrophy; no keloid), regular (partially visible scar; moderate atrophy; visible keloid), or poor (totally visible scar; craniofacial deformity), similar to previous report from our group.[12] Temporal muscle atrophy was evaluated by a similar method as grade I (mild, flattening of the temporal musculature), grade II (moderate, clear atrophy of the temporal musculature), or grade III (severe, deformity of the craniofacial contour) in the last clinic visit between 12 and 18 months after the surgery. Interobserver κ scores were >0.72 in all evaluations.

Bold indicates statistically significant findings after multiple comparisons correction ($P < 0.0016$).

Continues

Table 4. Continued

Outcome	Transeyelid Approach Group	Nanopterional Approach Group	P (Transeyelid Approach × Nanopterional Approach)	Pterional Approach Group	P (Minimally Invasive Surgery × Pterional Approach)
Facial nerve palsy					
15–30 days	0	2 (5.88)	0.44	14 (34.14)	**<0.001**
3–6 months	0	0	—	6 (14.63)	0.0042
12–18 months	0	0	—	4 (9.75)	0.032
Residual aneurysm neck					
≤2 mm	4 (11.11)	2 (5.88)	0.72	4 (9.75)	0.82
>2 mm	0	0		1 (2.43)	

Values are number (%) except where indicated otherwise.
Aesthetic results: excellent (no visible scar; no atrophy; no keloid), good (partially visible scar; mild atrophy; no keloid), regular (partially visible scar; moderate atrophy; visible keloid), or poor (totally visible scar; craniofacial deformity), similar to previous report from our group.[12] Temporal muscle atrophy was evaluated by a similar method as grade I (mild, flattening of the temporal musculature), grade II (moderate, clear atrophy of the temporal musculature), or grade III (severe, deformity of the craniofacial contour) in the last clinic visit between 12 and 18 months after the surgery. Interobserver κ scores were >0.72 in all evaluations.
Bold indicates statistically significant findings after multiple comparisons correction ($P < 0.0016$).

MIS and PtA groups in terms of surgical safety. Comparisons between the TelA group and NPtA group showed no statistically significant differences in all primary outcome variables (Table 4).

Secondary Outcome

Shorter operative times were observed in the MIS group compared with the PtA group ($P < 0.001$) but did not differ significantly between MIS subgroups ($P = 0.06$). Both temporary and permanent postoperative facial nerve palsy were more frequent in the PtA group ($n = 14$ [35%] temporary and $n = 4$ [10%] permanent) compared with the MIS group (2 patients in the NPtA group [5.88%] had temporary palsy but no patient in the MIS group had permanent palsy), but this finding did not reach statistical significance after Bonferroni correction ($P = 0.032$). There was no difference in the rate of facial nerve palsy between MIS subgroups ($P = 0.44$) (Table 4).

Radiologic outcomes were assessed at 3–6 months after the surgery. A total of 108 patients underwent postoperative digital subtraction angiography. In total, 11 (9.9%) residual aneurysms were observed after clipping, with 4 occurring in the TelA group (11.43%), and 2 (5.88%) and 5 (12.19%) in the NPtA and PtA groups, respectively. Only 1 patient in the PtA group with a complex middle cerebral aneurysm had a significant residual requiring reoperation. All other residual aneurysms, which measured between 1 and 2 mm, were followed and no recurrence was noted. There were no significant differences between groups or subgroups.

Aesthetic Results

MIS patients reported higher satisfaction scores related to their cosmetic results at all 3 time points of evaluation compared with PtA group patients ($P < 0.001$) (Table 5). Although good satisfaction (VAS score >7) was reported by all MIS group patients, TelA group patients reported higher scores than did NPtA group patients at last clinic visit (27 Tel group patients vs. 14 NPtA group patients had 10 VAS scores; $P = 0.0096$), but this did not reach statistical significance after multiple comparisons correction. These results were in keeping with the independent observers' evaluation.

Temporal muscle atrophy was more frequent and severe in the PtA group. Of patients in the PtA group, 35% presented some degree of muscle atrophy compared with 12.8% of MIS patients, and only PtA group patients ended up having severe (grade III) temporal muscle atrophy (10%) ($P = 0.0034$), nearing statistical significance. Although less frequent and less severe with MIS, there was no difference between MIS subgroups in terms of muscle wasting, with 2 (5.5%) TelA versus 7 (20%) NPtA, respectively ($P = 0.068$).

Patient Satisfaction

MIS patients reported consistently higher satisfaction in all 5 categories during all time points of evaluation. Masticatory alterations were more frequent and severe in the NPtA (64.7%; mean VAS score, 3.05 ± 2.66) and PtA groups (80%; mean VAS score, 4.45 ± 2.95) during the first clinic visit compared with TleA (16.67%; mean VAS score, 0.7 ± 1.78). Although less pronounced, this gap was still present at last evaluation when no TleA patients reported any degree of masticatory alteration compared with 5.88% of NPtA and 27.5% of PtA patients. Differences between MIS and PtA were significant ($P < 0.001$), but differences between MIS subgroups were significant only during the first evaluation ($P < 0.001$). Medium-term and long-term patient satisfaction scores did not significantly differ between groups ($P = 0.07$ and $P = 0.14$, respectively).

PtA patients also had significantly more facial numbness and discomfort compared with MIS patients ($P < 0.001$). Similar findings were reported regarding headache and local pain because

Table 5. Satisfaction and Quality of Life

	First Clinic Visit (15–30 days)					Second Clinic Visit (3–6 months)					Third Clinic Visit (12–18 months)				
Means	**TelA**	**NPtA**	***P* TleA × NPtA**	**PtA**	***P* MIS × PtA**	**TelA**	**NPtA**	***P* TleA × NPtA**	**PtA**	***P* MIS × PtA**	**TelA**	**NPtA**	***P* TleA × NPtA**	**PtA**	***P* MIS × PtA**
Patient satisfaction self-evaluation															
Masticatory alterations															
Mean score	0.7 (1.78)	3.05 (2.66)	**<0.001**	4.45 (2.95)	**<0.001**	0.26 (0.88)	0.91 (1.83)	0.074	2.05 (2.51)	**<0.001**	0	0.2 (0.84)	0.148	1.05 (1.79)	**<0.001**
Patients, n (%)	6 (16.67)	22 (64.71)		32 (80)		3 (8.33)	8 (23.53)		17 (42.5)		0	2 (5.88)		11 (27.55)	
Facial numbness															
Mean score	1.41 (1.84)	2.17 (2.65)	0.23	2.9 (1.99)	**<0.001**	0.44 (1.23)	0.32 (0.94)	0.98	0.82 (1.29)	**<0.001**	0.17 (1.02)	0.05 (0.34)	0.61	0.25 (0.89)	**<0.001**
Patients, n (%)	14 (38.89)	8 (23.53)		30 (75)		8 (22.2)	7 (20.59)		23 (57.5)		2 (5.56)	1 (2.94)		14 (35)	
Headache/local pain															
Mean score	1.41 (1.84)	2.17 (2.65)	0.24	2.9 (1.99)	**<0.001**	0.44 (1.23)	0.32 (0.94)	0.80	0.82 (1.29)	0.025	0.17 (1.02)	0.05 (0.34)	0.99	0.25 (0.89)	0.267
Patients, n (%)	15 (41.67)	16 (47.06)		38 (95)		5 (13.89)	4 (11.76)		12 (30)		1 (2.78)	1 (2.94)		3 (7.5)	
Cosmetic results	9.2 (0.74)	8.67 (1.09)	0.0075	5.75 (1.49)	**<0.001**	9.64 (0.54)	9.09 (0.94)	0.0015	6.75 (1.51)	**<0.001**	9.67 (0.58)	9.14 (0.85)	0.0096	6.85 (1.14)	**<0.001**
Overall satisfaction	8.61 (1.20)	8.64 (0.77)	0.55	6.12 (1.11)	**<0.001**	9.13 (0.86)	8.76 (0.60)	0.0061	6.67 (1.09)	**<0.001**	9.02 (0.77)	8.61 (0.65)	0.0095	6.70 (0.88)	**<0.001**
Quality of life self-evaluation															
Physical health	16.82 (2.27)	17.09 (1.84)	0.51	13.98 (1.89)	**<0.001**	17.97 (1.75)	17.93 (1.52)	0.73	16.22 (2.41)	**<0.001**	18.57 (1.15)	18.34 (1.19)	0.30	17.19 (1.70)	**<0.001**
Psycho logical	17.39 (1.61)	17.29 (2.46)	0.67	14.41 (2.08)	**<0.001**	18.48 (1.24)	18.27 (1.38)	0.68	16.31 (1.70)	**<0.001**	18.84 (1.14)	18.72 (1.33)	0.74	16.68 (1.88)	**<0.001**
Social relations	15.9 (2.31)	16.16 (2.22)	0.54	14.22 (2.51)	**<0.001**	17.02 (2.56)	16.78 (2)	0.47	15.54 (1.6)	**<0.001**	16.75 (2.08)	16.63 (1.89)	0.96	15.38 (1.56)	**<0.001**
Environment	15.68 (1.73)	14.87 (1.98)	0.18	13.29 (1.72)	**<0.001**	16.24 (1.84)	15.47 (1.65)	0.11	14.11 (1.36)	**<0.001**	16.16 (1.59)	15.69 (1.26)	0.44	14.58 (1.24)	**<0.001**
Self-evaluation	16 (3.19)	16.59 (2.63)	0.48	12.36 (2.47)	**<0.001**	17.76 (2.5)	17.71 (2.26)	0.89	15.23 (2.70)	**<0.001**	18.35 (2.17)	18.29 (2.37)	0.98	16.21 (2.28)	**<0.001**
Total	16.43 (1.69)	16.31 (1.66)	0.96	13.77 (1.45)	**<0.001**	17.43 (1.4)	17.1 (1.26)	0.41	15.43 (1.61)	**<0.001**	17.67 (1.14)	17.42 (0.97)	0.45	15.99 (1.32)	**<0.001**

In the first 3 aspects (masticatory alterations, facial numbness, headache/local pain), patients scored from 0 (without any disturbance) to 10 (worst discomfort imaginable). Visual analog scale was also used for aesthetic and general evaluation of the procedure: scores varied from 0 (totally dissatisfied) to 10 (totally satisfied). Quality of life was assessed by the World Health Organization Quality of Life Abbreviated questionnaire. Three patients were excluded from the quality-of-life analysis because they were diagnosed with depression (1 patient with borderline and 1 with severe depression in the TelA group, and 1 patient with moderate depression in the PtA group). No patients in the NPtA group were excluded.

Bold indicates statistically significant findings after multiple comparisons correction ($P < 0.0016$).

TelA, transeyelid approach; NPtA, nanopterional approach; PtA, pterional approach; MIS, minimally invasive surgery.

there were significantly lower rates reported in the MIS versus PtA groups ($P < 0.001$), but the results were not significant between MIS subgroups ($P = 0.83$). Mid-term and long-term rates did not differ between groups and subgroups (Table 5).

Overall procedure satisfaction was also significantly higher in the MIS than in the PtA group ($P < 0.001$ at all evaluation time points) (Table 5). Although comparison between MIS subgroups showed no difference at first evaluation ($P = 0.55$), TelA patients were overall more satisfied with the procedure than were NPtA patients at mid-term and long-term follow-up ($P = 0.006$ and 0.009, respectively).

Quality of Life

All quality-of-life-domain mean scores were statistically higher in the MIS compared with PtA group at all 3 evaluation time points ($P < 0.001$). On the other hand, comparisons of MIS subgroup domain mean scores did not show significant difference (Table 5). No cutoff point for "good" or "poor" quality of life was used in the current analysis because of subjective interpretations.[16,17]

DISCUSSION

The present study shows that the use of MIS open techniques for treatment of saccular UIAs of the anterior circulation is equally safe and effective compared with standard open surgery but without clear superiority between the 2 investigated MIS techniques. Nevertheless, MIS techniques showed superior cosmetic, satisfaction, and quality-of-life outcomes with shorter operative time, suggesting that MIS techniques can be implemented in certain situations to reduce surgical trauma without sacrificing patient safety or treatment efficacy. However, these conclusions cannot be extended to the surgery of ruptured intracranial aneurysms, in which other risks may be present.

The present cohort consists of a uniform population of patients and aneurysms in terms of demographics and aneurysm characteristics. The 2 groups (MIS vs. PtA) and the 2 subgroups (TelA vs. NPtA) were also homogeneous. Therefore, we infer that adequate randomization was achieved, and valuable conclusions on comparisons of surgical, clinical, aesthetic, and quality-of-life outcomes can be made.

Primary Outcomes

Although most studies consider an mRS score >2 as the cutoff for poor prognosis,[18-20] the present study used a stricter criterion because most patients were asymptomatic before surgery. Therefore, we understood that any postoperative increase in mRS score in this population would be considered unacceptable. However, the only conclusion that can be made is that MIS is equally as safe and effective in the treatment of UIA of the anterior circulation compared with standard open surgery in terms of mortality and neurologic outcome. The rates of primary outcomes in the present study are consistent with reports from previously reported series, with a mortality of 0%–1.6% and morbidity of 6.2%–9.4% for the treatment of unruptured aneurysms.[20,21]

Surgical safety as defined by the measured subitems showed conflicting results, but no statistical differences were noted. The homogeneity of intraoperative events reinforces the uniformity of the groups in relation to the location, size, and complexity of the aneurysms. All types of aneurysms were clipped by both MIS approaches without specific challenges. However, the endoscope played an important role in identifying perforating arteries and verifying proper clipping in anterior communicating artery and internal carotid artery (posterior communicating artery and choroidal segments) aneurysms because the angle of vision provided by the microscope is limited in the TelA and NPtA. Postoperative events also showed similar results. Again, as with mortality, no clear superiority was found but the safety of MIS was equivalent to standard techniques.

Secondary Outcomes

The operation time of MIS was on average 30% shorter than standard surgery. The literature supports shorter operative times with an average of 70–160 minutes in MIS versus 120–240 minutes in standard open surgeries.[22,23] Moreover, a recent study reported by Dasenbrock et al.[19] of the surgical results of 626 patients with UIAs found operative time >330 minutes was associated with increased complications. In our cohort, only the PtA group had surgeries >330 minutes (21%).

Shorter operative times observed in the MIS group did not sacrifice surgical efficacy. In the literature, numbers are scarce but Raftopoulos et al.[24] reported a total aneurysm occlusion rate of 93.2%. Spetzler et al.[25] reported a total aneurysm obliteration rate of 85% (ruptured and UIAs). In the present study, 5 patients in the PtA group (12.1%) and 6 in the MIS group (8.5%) had residual aneurysmal necks. The use of the endoscope for adjunctive inspection in the MIS group may have contributed to the lower rates of residual aneurysms. Nevertheless, most residual aneurysm necks were considered clinically insignificant without a need for subsequent reoperation and without development of delayed enlargement or hemorrhage.

In addition, the data of the present study also indicate that early discharge is safe, similar to the literature on patients undergoing craniotomies.[26,27] Discharge after 24 hours of observation is therefore a safe strategy in the management of UIAs. The results of these secondary outcomes suggest that MIS approaches are as efficient and effective as standard approaches in clipping of UIAs.

Patient Satisfaction and Quality of Life

Our results show a clear superiority of the MIS over the PtA group in all aspects of patient satisfaction and quality-of-life evaluations. No differences in these same measurements were identified between MIS subgroups. One potential confounder was the lack of haircut standardization. This difference could influence the interpretation of aesthetic results, especially at early outpatient return and with self-reported satisfaction. Despite these limitations, the results of independent observer evaluations were congruent with self-reported outcomes and only patients in the PtA group had poor aesthetic outcomes and grade III temporal muscle atrophy.

In patients undergoing pterional craniotomy, 80%–91% report orofacial pain when evaluated during the first postoperative month.[28] In the present study, 73.1% of patients reported orofacial pain at the first clinic visit, decreasing to 9.7% by last follow-up. Accordingly, differences in pain outcomes were not observed between groups during late follow-up visits because the wounds have been fully healed. With use of MIS techniques such as the supraorbital craniotomy, Reisch et al.[29] reported that only 23% patients experienced orofacial pain at early follow-up. Prospective data from the present study show similar or better rates, with 12.8% and 2.8% of orofacial pain at mid and late follow-up, respectively.

We found improved self-reported WHOQOL scores in patients undergoing MIS compared with the standard PtA approach. We chose the WHOQOL for assessing quality of life in the present study to prevent the ceiling effect, which could have homogenized the study groups, especially at long-term follow-up. Although higher quality-of-life scores were observed in the MIS group, it cannot be assumed that patients in the PtA group had poor quality of life after treatment, because cutoff scores are not reliable.[16,17]

CONCLUSIONS

TleA and NPtA are safe and effective yet less invasive alternatives for the treatment of UIAs of the anterior circulation compared with PtA. MIS is superior to the PtA in achieving excellent functional, cosmetic, and quality-of-life outcomes. TelA and NPtA can be considered equivalents in terms of the outcomes measured in the present study.

CRediT AUTHORSHIP CONTRIBUTION STATEMENT

Mauricio Mandel: Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing - original draft, Validation. **Rafael Tutihashi:** Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing - original draft, Validation. **Yiping Li:** Writing - review & editing. **Jefferson Rosi:** Investigation. **Brasil Chian Ping Jeng:** Investigation. **Manoel Jacobsen Teixeira:** Supervision, Funding acquisition. **Eberval Gadelha Figueiredo:** Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing - original draft, Validation, Supervision.

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Conflict of interest statement: The authors declare that the article content was composed in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest. Funding for this study was received from Hospital das Clinicas of University of Sao Paulo Medical School.

Received 18 March 2021; accepted 17 April 2021

Citation: World Neurosurg. (2021) 151:e533-e544.
https://doi.org/10.1016/j.wneu.2021.04.079

Journal homepage: ***www.journals.elsevier.com/world-neurosurgery***

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